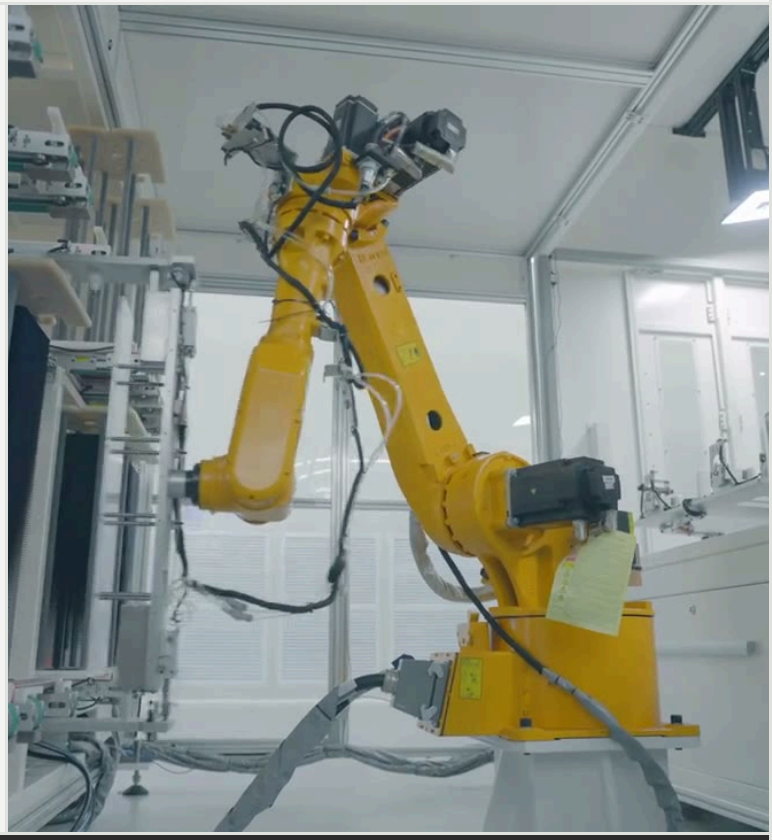
The Great Indian Engineers

Theory, embodied.

One accountable team, from first sketch to shipped hardware.

Most ideas die after the breakthrough. In the board that won't hold a signal, the company nobody has time to form, the raise nobody prepped for. We carry all of it. You keep the idea.

- Free scoping call. Your idea stays yours from day one.



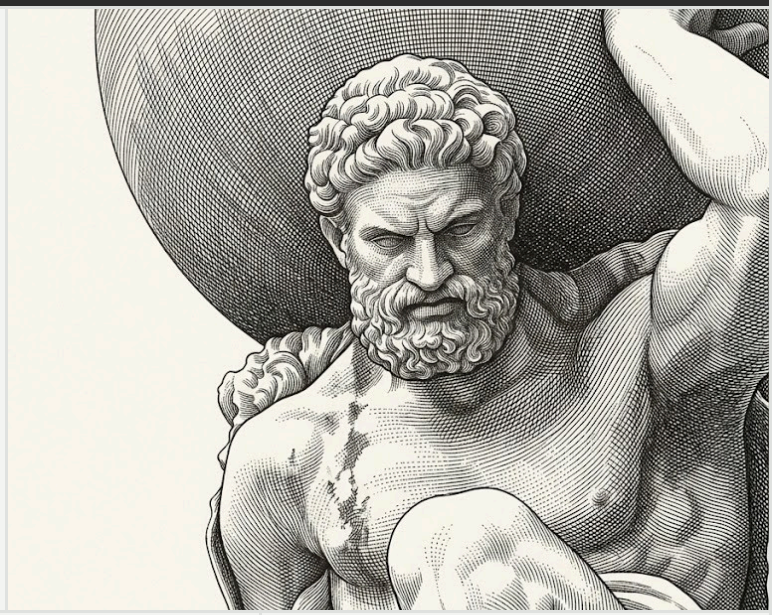
INSTITUTIONAL RIGOR · MEDICAL-GRADE DISCIPLINE · ISO 9001:2015 FABRICATION

FreeRTOS | STM32CubeMX | BLE/GATT | Zero Data-Packet Loss | DFM-Ready Gerber

Our conviction

You carry the vision. We carry the weight.

The discovery is yours: the thesis, the data, the IP. What we carry is the operational weight of turning an idea into something you can hold, defend, publish, and sell.



PRINCIPLE I

An idea is only real once it has a form

Physical, legal, and commercial form is what makes it exist.

PRINCIPLE II

The weight shouldn't fall on the creator

We shoulder the operational weight so the thinking stays yours.

PRINCIPLE III

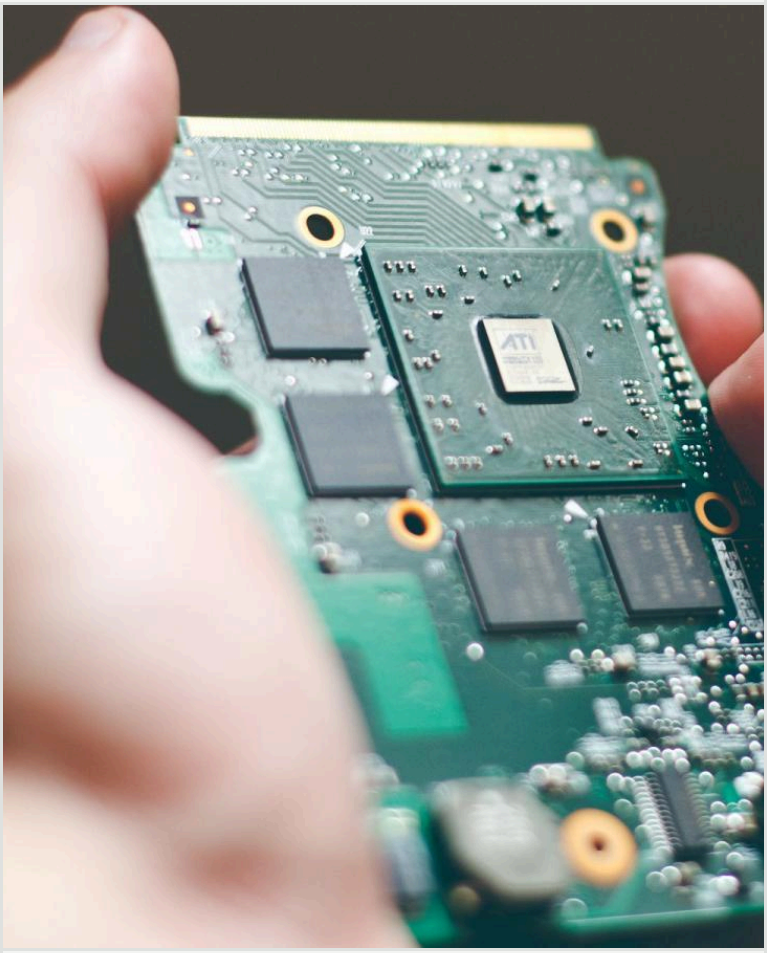
A form we build, you own completely

Files, IP, entity, brand. All of it, handed back. Zero lock-in.

Four pillars, one roof

Engineering, products, business, and heavy fabrication.

Not a menu of services. Each pillar gives your idea a different form, and lifts a different weight. All held by one accountable team.



PILLAR A

Embedded Hardware & Engineering

Off-the-shelf boards fail exactly where research gets interesting. We build hardware that survives the edge cases your data depends on.

01

Custom PCB & schematic

Up to 6 layers, DFM-ready Gerber

02

Embedded firmware

Bare-metal, FreeRTOS, BLE/GATT

03

3D enclosures

Designed with the PCB footprint

04

Edge AI / TinyML

On-device inference, no cloud



PILLAR B

Additive Manufacturing & Product Lines

Additive manufacturing, from a bespoke commission to a scalable retail line.

01

Industrial enclosures

PA-CF, PET-CF ruggedized housings

02

Kinetic IoT & decor

Motorized, connected desktop art

03

Automotive plastics

Discontinued & custom cabin parts

04

Retail & e-commerce

High-margin Amazon / D2C runs

PILLAR C

Business Enablement & Advisory

A prototype is not a company. We build the company too.

01

Branding & marketing

Identity, Amazon, D2C content

02

Startup services

Incorporation, compliance, and fundraising

03

Fundraising & GTM

DST-NIDHI, BIRAC BIG, MeitY, angels

04

Audit & operations

Compliance, BOM traceability, SOPs





PILLAR D

Heavy Fabrication & Manufacturing

One Purchase Order, from RFQ to a delivered, certified structure.

01

Single-source accountability

RFQ to logistics, one PO

02

In-house machinery

Laser, plasma, 320-ton press brake

03

Certified welding

GMAW (MIG) and SMAW (ARC)

04

QA & documentation

ISO 9001:2015, mill certs, WPS

Who we serve

Three audiences, one accountable roof.

01

Research labs & PhD scholars

Field-ready custom hardware, calibrated, documented, and yours to publish.

02

Founders & academic spin-offs

A validated prototype taken through fabrication, branding, incorporation, and

03

Industrial & OEM clients

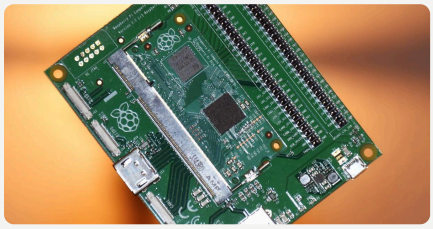
Heavy structural fabrication at scale, under single-source accountability.

Core technical stack

Capabilities built for research and product teams.

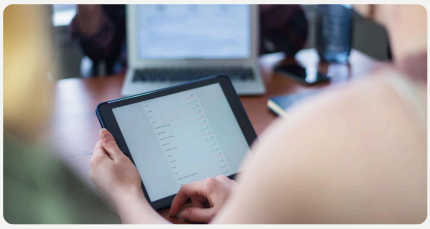
Firmware & silicon mastery

Bare-metal and FreeRTOS across Espressif ESP-IDF, Nordic nRF Connect, and STM32.



Full-stack data visualization

Cross-platform app interfaces and desktop diagnostic dashboards that read telemetry in real time.



Edge AI & on-device ML

Lightweight models on microcontrollers via Edge Impulse and TensorFlow Lite.



Engagement model

One phased path from concept to launch.

25%

Phase 1

Proof of Concept

Dev-board breadboarding, sensor communication, logic validation.

You get: proof the idea actually works.

50%

Phase 2

Engineering Validation

Custom PCB, miniaturisation, 3D-printed enclosure, bare-metal firmware.

You get: a working prototype you can hold.

75%

Phase 3

Design Validation

Fully integrated device, optimized firmware, diagnostic GUI.

You get: field-tested hardware, ready for real data.

100%

Phase 4

Launch Readiness

Small-batch fabrication, branding, marketplace listing, compliance.

You get: a sellable, fundable product.

CONTACT US

You dream it. We'll carry it.

Take the operational weight off your plate. One accountable team spanning engineering, legal, branding, and market.

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www.tgie.in

Thirumudivakkam, Chennai · India

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